

AKQI, China

WMO: 517110

Lat: 40.9414N

Lon: 78.4512E

Elev: 6516

StdP: 11.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	2.7	-10.9	3.9	19.1	-7.5	4.7	16.7	13.2	22.7	12.0	21.7	3.8	250	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	83.7	57.2	81.1	56.2	78.7	55.4	60.5	76.4	59.1	74.7	57.9	73.1	4.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	83.7	65.3	53.7	78.3	64.2	52.0	73.6	63.6	30.0	76.2	29.0	74.9	28.0	72.9	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	11.2	9.9	-3.9	89.9	4.6	2.3	-7.3	91.6	-10.0	93.0	-12.6	94.3	-16.0	96.0		
			-4.6	63.4	4.5	2.6	-7.9	65.3	-10.5	66.8	-13.0	68.3	-16.3	70.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	16.9	23.8	38.5	51.0	58.5	64.5	67.8	66.2	58.8	47.5	34.3	21.5	(6)
(7)		DBStd	18.77	7.92	7.71	7.99	7.58	6.06	4.82	4.73	4.76	5.57	6.35	7.14	7.48	(7)
(8)		HDD50	3693	1027	733	363	77	10	0	0	0	5	125	471	882	(8)
(9)		HDD65	7240	1492	1153	823	422	211	67	24	43	196	542	920	1347	(9)
(10)		CDD50	2188	0	0	6	106	272	434	552	501	268	49	0	0	(10)
(11)		CDD65	261	0	0	0	1	9	50	112	80	9	0	0	0	(11)
(12)		CDH74	1972	0	0	0	20	89	385	849	553	75	1	0	0	(12)
(13)		CDH80	348	0	0	0	1	3	51	191	97	5	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.0	5.2	5.5	5.7	5.2	5.3	5.0	4.9	4.7	4.7	5.0	4.9	(14)
(15)	Precipitation	PrecAvg	7.4	0.1	0.2	0.4	0.6	1.0	1.1	1.3	1.5	0.9	0.1	0.1	0.0	(15)
(16)		PrecMax	13.8	0.3	0.9	1.7	2.4	2.6	3.4	3.9	4.1	2.6	0.9	0.6	0.6	(16)
(17)		PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.1	0.2	0.5	0.6	0.7	1.0	0.9	1.1	0.6	0.2	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	47.6	64.0	76.8	79.6	84.4	88.6	86.4	80.1	70.3	57.2	46.4	(19)
MCWB			29.8	32.9	42.2	49.1	51.5	55.5	58.6	58.8	54.8	48.6	40.2	32.8	(20)	
2%		DB	36.2	42.0	59.2	72.0	76.6	81.2	84.7	82.7	75.9	65.5	52.3	39.6	(21)	
		MCWB	26.1	30.2	39.7	47.3	50.7	54.6	57.8	57.5	52.9	46.5	37.0	28.7	(22)	
5%		DB	31.6	38.2	55.2	68.4	73.9	78.6	81.8	79.8	72.7	62.0	48.6	35.1	(23)	
		MCWB	23.4	28.4	38.3	45.8	49.8	54.1	57.1	56.8	52.4	44.9	35.8	26.0	(24)	
10%		DB	27.6	34.9	51.3	64.3	70.9	75.7	78.9	77.0	69.6	58.4	44.7	31.5	(25)	
		MCWB	21.0	26.7	36.5	43.8	49.0	53.5	56.5	56.3	51.4	43.2	33.8	24.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.5	33.5	43.2	50.8	54.6	59.7	62.8	62.3	57.4	50.1	40.6	32.8	(27)
MDCB			42.0	45.0	61.2	72.4	71.9	77.5	79.5	77.1	73.5	65.1	55.1	45.0	(28)	
2%		WB	26.4	30.8	40.8	48.5	52.8	57.4	60.6	60.7	55.4	48.0	38.3	28.9	(29)	
		MDCB	35.5	40.6	56.2	68.3	71.1	73.7	77.2	75.9	70.3	62.3	50.2	38.8	(30)	
5%		WB	23.7	28.8	39.1	46.8	51.5	56.1	59.3	59.2	54.0	46.4	36.3	26.5	(31)	
		MDCB	31.3	37.2	53.1	65.3	69.1	72.0	75.4	74.1	67.9	59.0	46.9	34.2	(32)	
10%		WB	21.1	27.0	37.4	45.1	50.4	55.0	58.2	58.0	52.6	44.6	34.5	24.2	(33)	
		MDCB	27.3	34.1	49.9	61.8	66.8	70.6	73.8	72.4	65.9	56.0	43.6	31.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.2	16.5	17.4	19.2	19.4	19.2	20.0	19.5	19.2	18.8	16.2	16.4	(35)	
5% DB		MCDBR	21.0	19.5	22.3	24.4	23.6	23.4	24.1	23.5	22.8	22.0	20.8	20.4	(36)	
		MCWBR	13.5	11.4	10.6	10.0	8.5	7.4	7.4	7.6	7.9	9.0	10.4	12.4	(37)	
		MCDBR	20.5	18.2	20.0	22.6	20.7	19.8	20.9	20.1	20.5	19.9	19.3	19.7	(38)	
5% WB		MCWBR	13.4	10.9	9.8	9.6	7.7	7.2	7.5	7.6	8.0	8.4	10.0	12.1	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.347	0.456	0.530	0.543	0.533	0.495	0.535	0.507	0.416	0.339	0.290	(40)	
(41)		taud	2.227	2.075	1.794	1.662	1.691	1.762	1.884	1.773	1.781	1.937	2.121	2.258	(41)	
(42)		Ebn at Noon	269	267	246	238	239	241	249	234	228	237	246	261	(42)	
(43)		Edh at Noon	30	41	62	76	75	70	62	67	62	47	33	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	757	1012	1361	1711	1944	2005	2001	1800	1513	1188	836	665	(44)	
(45)		RadStd	65	87	106	106	124	115	114	94	99	70	56	60	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	-0.86	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)		N/A	N/A	N/A	N/A	-1.08	N/A	N/A	N/A	N/A

Nomenclature: See separate page